Electrical Circuit-II
ECE-1201

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“Read! In the name of thy Lord and Cherisher, Who created-”.

[Sura Al-’Alaq]

Study Materials

- **Fundamental of Electric Circuits**

  Charles K. Alexander, Matthew N. O. Sadiku

- **Introductory Circuit Analysis**

  Robert L. Boylestad

- **Alternating Current Circuits**

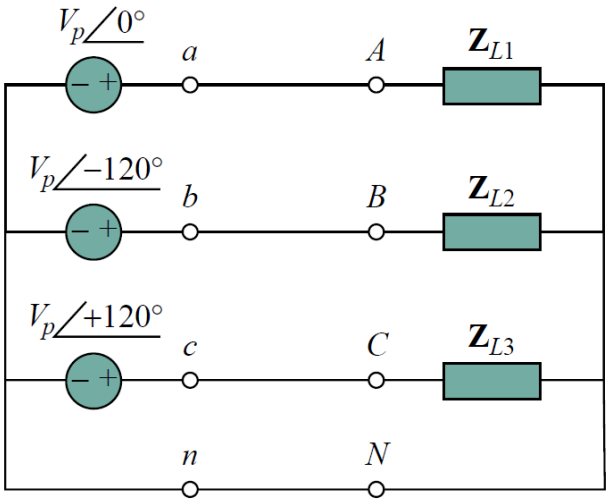
  Russell M Kerchner, George F. Corcoran

Polyphase

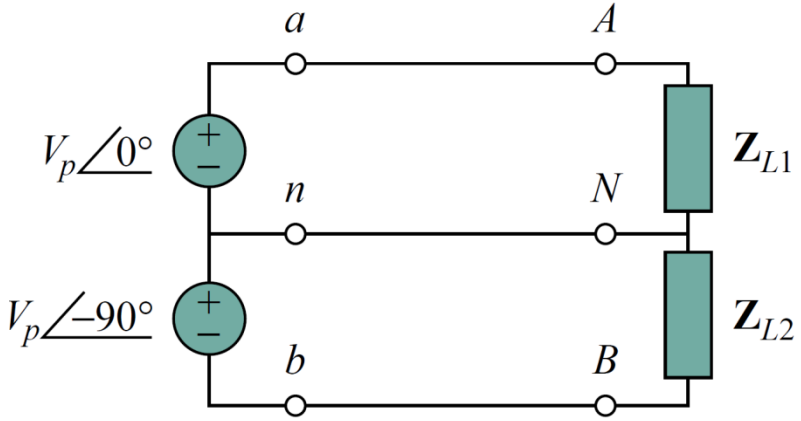
Circuits or systems in which the ac sources operate at the same frequency but different phases are known as polyphase

A **polyphase** system is simply **several single-phase** systems which are displaced in different phases from one another. The single-phase systems which form the polyphase systems are generally **interconnected** in some way.

Polyphase



A **three-phase** four wire system



A **two-phase** three wire system

Three-phase Systems

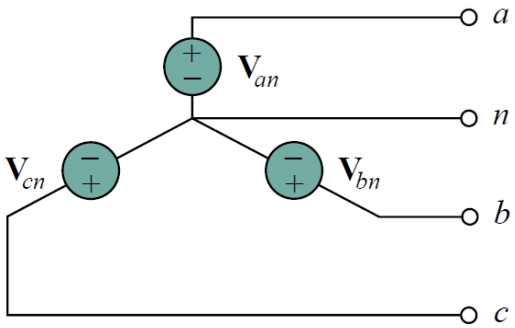
Three-phase systems are important for the following **reasons.**

- **First,** nearly all electric power is generated and distributed in three-phase at the operating frequency of 60 Hz or 50 Hz.
- **Second,** the instantaneous power in a three-phase system can be constant (not pulsating).
- **Third,** for the same amount of power, the three-phase system is more economical than the single-phase, i.e., the amount of wire required for a three-phase system is less than that required for an equivalent single-phase system.
- **Fourth,** in general, most larger motors are three phase because they are essentially self-starting and do not require a special design or additional starting circuitry.

Balanced Three-phase Systems

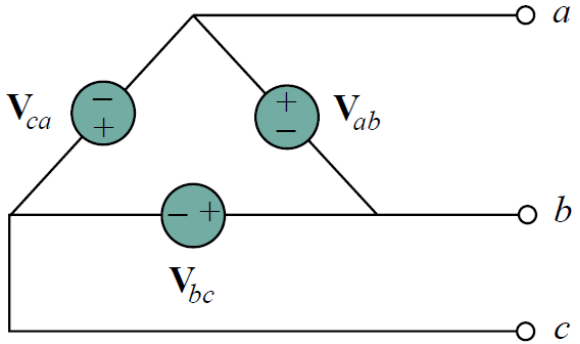
A balanced three-phase system is one in which the load impedances of all the phases are equal when the sources supply equal voltages.

A typical three-phase system consists of **three voltage** sources (three-phase current sources are very scarce) connected to **loads by three or four wires** (or transmission lines). A three-phase system is equivalent to three single-phase circuits. The voltage sources can be either **wye-**connected or **delta** connected.



a

Three phase voltage sources: (a) Y-connected (b) Δ connected

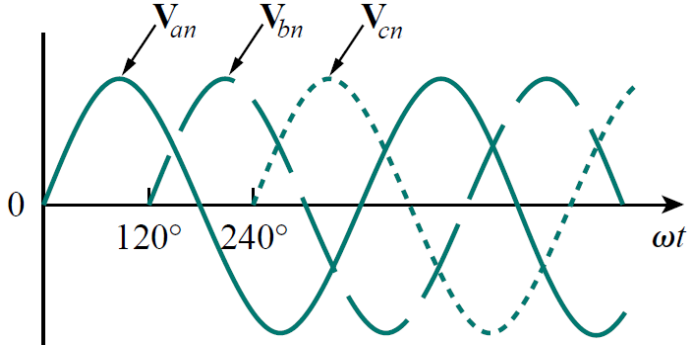


b

Representing Three-phase Systems

Sinusoidal expression

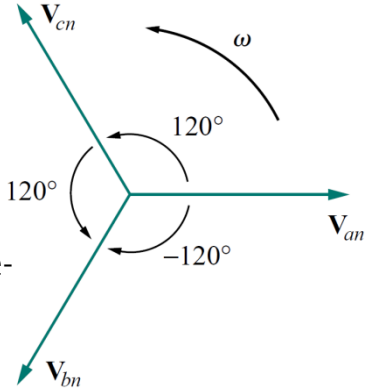
$$v_{an} = V_m \sin(\omega t)$$
$$v_{bn} = V_m \sin(\omega t - 120°)$$
$$v_{cn} = V_m \sin(\omega t - 120° - 120°)$$
$$= V_m \sin(\omega t - 240°)$$
$$= V_m \sin(\omega t + 120°)$$



Phase voltages of a three-phase voltage source. The generated voltages are 120° apart from each other.

Phasor expression

$$V_{an} = \frac{V_m}{\sqrt{2}} \sin(\omega t)$$
$$V_{bn} = \frac{V_m}{\sqrt{2}} \sin(\omega t - 120°)$$
$$V_{cn} = \frac{V_m}{\sqrt{2}} \sin(\omega t + 120°)$$



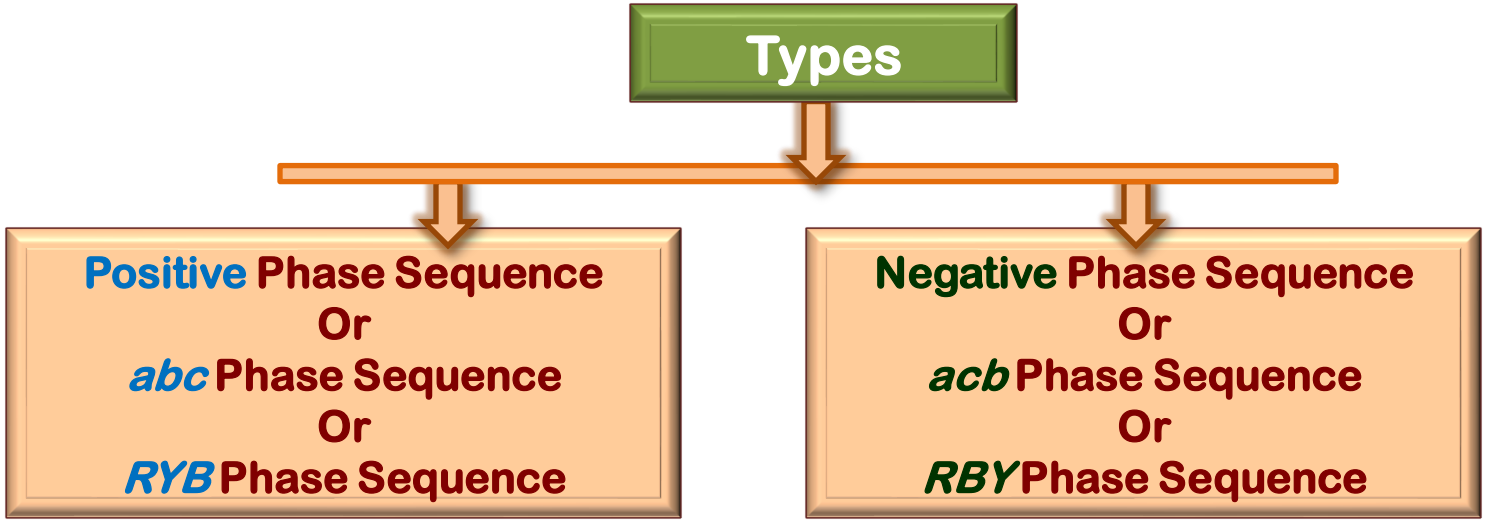
Phasor diagram for the above three-phase voltages.

Phase Sequence

Phase sequence is the time order in which the voltages pass through their respective maximum value

Or

It is the order in which the phase voltages reach their peak (or maximum) values with respect to time.



Determining Phase Sequence

The phase sequence can be determined by the order in which the **phasors** representing the phase voltages pass through a **fixed point** on the phasor diagram if the phasors are rotated in a **counterclockwise** direction.

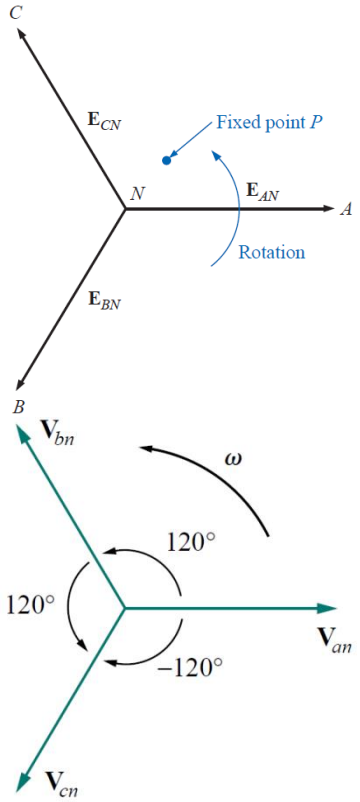
Important Conclusion
For determining phase sequence

- Have to convert into **phasor** form.
- Have to assume a **fixed point.**
- Have to rotate in **counterclockwise.**

Determining Phase Sequence

For the given figure, the phase sequence is *abc*. However, since the fixed point can be chosen anywhere on the phasor diagram, the sequence can also be written as bca or cab

Positive Phase Sequence
Or
abc Phase Sequence $= \begin{cases} abc \\ bca \\ cab \end{cases}$
Or
***RYB* Phase Sequence**



For the given figure, the phase sequence is *acb*. However, since the fixed point can be chosen anywhere on the phasor diagram, the sequence can also be written as *cba* or *bac*.

Negative Phase Sequence
Or
acb Phase Sequence $= \begin{cases} acb \\ cba \\ bac \end{cases}$
Or
***RBY* Phase Sequence**

Determining Phase Sequence

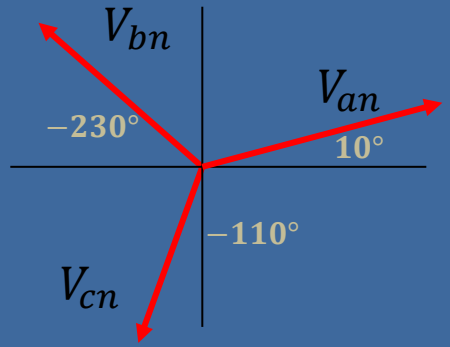
Example 12.1 [of the book written by *"Alexander"*]:

Solution:

$$V_{an} = \frac{200}{\sqrt{2}}\cos(10°)$$

$$V_{bn} = \frac{200}{\sqrt{2}}\cos(-230°)$$

$$V_{cn} = \frac{200}{\sqrt{2}}\cos(-110°)$$



Practice Problem 12.1 of Alexander and similar types